Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 02
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. If two positive integers p and q can be expressed as $p = ab^2$ and $q = a^3b$, where a, b are prime numbers, then LCM(p, q) is: [1]

(a) ab (b) a^2b^2 (c) a^3b^2 (d) a^4b^3

Q2. The product of a non-zero rational number and an irrational number is: [1]

(a) always rational (b) always irrational
(c) rational or irrational (d) one

Q3. The least number that is divisible by all the natural numbers from 1 to 10 (both inclusive) is: [1]

(a) 10 (b) 100 (c) 504 (d) 2520

Q4. If HCF$(26, 169) = 13$, then LCM$(26, 169)$ is: [1]

(a) 26 (b) 52 (c) 338 (d) 13

Q5. **Assertion (A):** The HCF of two numbers is 5 and their product is 150, then their LCM is 30.
Reason (R): For any two positive integers a and b, HCF$(a, b) \times$ LCM$(a, b) = a \times b$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Check whether 6^n can end with the digit 0 for any natural number $n \in \mathbb{N}$. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Prove that $\sqrt{7}$ is an irrational number. [3]

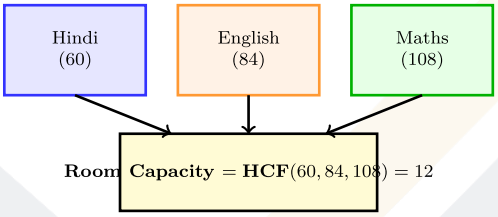
Q8. Prove that $2\sqrt{3} - 1$ is an irrational number. [3]

Q9. Prove that $\sqrt{2} + \sqrt{3}$ is an irrational number. [3]

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: CASE STUDY 02: NATIONAL EDUCATIONAL SEMINAR ROOM ALLOCATION [4 Marks]

The Ministry of Education is organising a 3-day National Seminar on Mathematical Sciences in New Delhi. The participating teachers are registered across three disciplines: Hindi (60 participants), English (84 participants), and Mathematics (108 participants). The event coordinator must allocate participants into parallel breakout rooms such that in each room, exactly the same number of participants are seated, and all participants in any given room belong to the same discipline.



(i) Find the maximum number of participants that can be accommodated in each breakout room. [2]

(ii) Calculate the minimum total number of breakout rooms required for the seminar. [1]

(iii) Express the Mathematics participant count (108) in canonical prime factorisation form. [1]

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